ZZ-B-220b
May 20, 1968
SUPERSEDING
Fed. Spec. ZZ-B-220a
April 21, 1960

FEDERAL SPECIFICATION

BELTING, V, ADJUSTABLE LINK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers adjustable link, V-belting for field replacement of V-belts.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Type. The belting shall be of one type.

1.2.2 Sizes. The belting shall be of the following sizes, as specified (see 6.2):

Size	Nominal width (inches) (at second hole from wide end of link)	Included angle (degrees) of belting to fit standard sheaves
00	5/16	34
0	3/8	34
A	1/2	36
B	21/32	38
C	7/8	38
D	1 1/4	38

FSC 3030

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:25Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-585 - Boxes; Wood, Wirebound.
PPP-B-591 - Boxes; Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes; Wood, Cleated-Plywood.
PPP-B-621 - Boxes; Wood Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications and Standards, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:25Z
Check the source to verify that this is the current version before use.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Classification:

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth St., N. W., Washington, D. C. 20036.)

Uniform Freight Classification or Consolidated Freight Classification:

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chidaco, Ill. 60606.)

3. REQUIREMENTS

3.1 Preproduction sample. The contractor shall furnish, within the time limit specified in the contract or order, a preproduction sample for inspection. This inspection, which shall consist of the examination and test specified in 4.3.1, shall be conducted prior to the submission for acceptance of the remainder of the belting furnished under the contract. The preproduction sample shall be identical to the belting the contractor proposes to furnish in fulfillment of the contract. Approval of the preproduction sample on the basis of the specified inspection shall not relieve the contractor of his obligation to meet the requirements of the specification for the remaining belting furnished under this contract.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of the type and quality generally recognized by industry as suitable for the intended purpose. Material shall be free from all defects and imperfections that may affect the serviceability or appearance of the finished product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:25Z
Check the source to verify that this is the current version before use.

3.2.1 Rubber. Rubber shall be either natural or synthetic compounds.

3.2.2 Fabric. Fabric shall be natural, synthetic, or a combination of natural and synthetic fabrics.

3.3 Standard product. Except as otherwise specified herein, the equipment and its component parts shall be a standard commercial product of the manufacturer or his suppliers. The parts shall be new and unused.

3.4 Interchangeability of items. All units of the same classification furnished under any specific contract shall be physically and mechanically identical.

3.5 Construction. The belting shall be of multiple-link construction assembled in lengths of 100 feet ± 2 inches. Links shall be constructed with interwoven or laminated fabric impregnated with rubber. The links shall be machine cut so that the included angle and width of the assembled links shall conform to the dimensions specified in 1.2.2. The links shall be joined by steel studs and washers to permit minimum length adjustment. The studs shall be cadmium-plated, zinc-coated, or japanned in accordance with the manufacturer's standard practice. Washers shall be fabricated of fabric, leather, ferrous, nonferrous, or synthetic material that will not degrade or limit the performance of the belt. The belting shall be coiled on metal or fiberboard reels in accordance with the manufacturer's standard practice.

3.6 Oil-resistance. The belting shall be oil-resistant. When tested as specified in 4.6.2, the width and thickness shall not be increased more than 10 percent, and neither the width nor the thickness shall show evidence of shrinkage.

3.7 Heat resistance. The belting shall be capable of withstanding a temperature of 158° F. No cracks or deterioration shall be evident when the belting is tested as specified in 4.6.3.

3.8 Static conductive and nonsparking. When specified (see 6.2), belting, except size D, shall be static conductive and nonsparking. In lieu of steel studs specified in 3.5, studs for nonsparking belts shall be fabricated of a nonsparking copper alloy. In addition, washers shall be fabricated of nonsparking, nonferrous materials. When tested as specified in 4.6.4, the maximum resistivity shall be 6 megohms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:25Z
Check the source to verify that this is the current version before use.

3.9 Identification marking. Each 100-foot reel shall be legibly marked with the appropriate size as specified in 1.2.2 and with the manufacturer's name or trademark. Reels shall be appropriately marked to indicate static conductive and nonsparking belting. Information marked on the reels shall also be marked on a tag attached to the exposed end of belting on each reel.

3.10 Assembly tools. The number of assembly tools to be furnished with each 100-foot reel of belting shall be as specified in the contract or order (see 6.2).

3.11 Assembly construction. The supplier shall furnish written instructions for assembly and disassembly of the links.

3.12 Workmanship. The quality of workmanship shall be such as to produce belting in accordance with the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. For purposes of inspection, a lot shall consist of all belting of the same size offered the Government at one time under a specific contract.

4.3 Classification of inspections The inspection requirements specified herein are classified as follows:

a. Preproduction inspection.
b. Quality conformance inspection.

4.3.1 Preproduction inspection. The preproduction sample shall consist of an adequate length of each size belting necessary to perform the examination of 4.5 and all tests of 4.6. Ten feet of the belting shall be used for the functional and durability test of 4.6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:25Z
Check the source to verify that this is the current version before use.

4.3.2 Quality Conformance inspection. Quality conformance inspection shall be performed on the samples selected in accordance with 4.4 from each lot offered to the Government. This inspection shall include the examination of 4.5. For inspection purposes, each sample shall be a reel containing 100 feet of belting.

4.4 Sampling for examination. A random sample of belting shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level II. The acceptable quality level (AQL) shall be 1.0 percent defective units for major defects and 2.5 percent defective units for minor defects.

4.5 Examination.

4.5.1 Visual examination. Samples selected in accordance with 4.4 shall be examined for the defects listed in table I. Any sample having one or more of the defects listed shall be considered a defective unit.

TABLE I. Classification of defects

Categories	Defects
Critical:	None defined.
Major:	
101	Size not as specified (see 1.2.2 and 3.5).
102	Material not as specified (see 3.2, 3.2.1, and 3.2.2).
103	Construction not as specified; components missing or damaged; links not machine cut to the included angle specified; studs not made of steel; washers not furnished; belting not coiled on reels; length of belting assembly not as specified (see 3.5).
104	Belting not oil-resistant; increase in width and thickness greater than 10 percent; evidence of width and thickness shrinkage (see 3.6).
105	Belting does not withstand temperature of 158° F.; evidence of cracks or deterioration (see 3.7).
106	Belting (except type D) not static conductive and nonsparking; studs not fabricated of a nonsparking copper alloy; washers not fabricated of nonsparking, nonferrous materials (see 3.8).
107	Identification marking not as specified (see 3.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:25Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of defects (con.)

Categories	Defects
108	Assembly tools not furnished; quantity not as specified (see 3.10).
109	Workmanship not as specified (see 3.12).
Minor:	
201	Instructions incomplete, missing, or illegible (see 3.11).

4.6 Tests.

4.6.1 Functional and durability test. The preproduction sample shall be assembled and tested for the period of time specified in table II. The belting shall operate adequately and shall show no evidence of failure.

TABLE II. Test requirements

Belt size	Nominal width (inches)	Test time (hours)
00	5/16	100
0	3/8	100
A	1/2	100
B	21/32	200
C	7/8	100
D	1 1/4	100

4.6.1.1 Test equipment. The test equipment shall be operated on a 230-volt, 60-cycle a.c. system and shall have the characteristics indicated in table III. The motor shall be in a fixed location and the generator shall be bolted to a frame mounted on ball-bearing wheels having free roll on a metal track. The drive shall be horizontal and the belt shall be run with the tight side on top. The load for each belt size shall be held constant during the test by use of a rheostat controlling an electrical resistance bridge. The weight for each belt size indicated in table III shall be suspended vertically from a wire cable supported on an idler pulley and fastened to the generator frame.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:25Z
Check the source to verify that this is the current version before use.

TABLE III. Test equipment characteristics

Belt size	Motor		Weight-tension (less generator) lbs.
	Hp.	Speed r.p.m.	
00	5	1750	80
0	5	1750	100
A	5	1750	100
B	5	1750	100
C	20	1750	200
D	20	1750	200

4.6.2 Oil-resistance test. The preproduction sample shall be immersed in ASTM oil No. 1 at a temperature of 158° F. for 22 hours. The width and thickness of the individual links shall be measured before and after immersion in the oil. Failure of any one link to meet the requirements of 3.6 shall be cause for rejection.

4.6.3 Heat-resistance test. The preproduction sample shall be suspended in an oven and heated at an ambient air temperature of 158° F. for 12 hours. Failure of the belting to meet the requirements of 3.7 shall be cause for rejection.

4.6.4 Static conductivity test. When static conductive belting is required (see 3.8), the preproduction sample shall be placed on 2 electrodes having a minimum diameter of 1/2-inch. The electrodes shall be spaced 8-1/2 inches between centers and shall be moistened with water to facilitate good conductivity. A force of 12-1/2 pounds shall be applied to each electrode to insure good contact between the belting and the electrodes. The resistivity (conductivity) between the electrodes shall be measured by means of an ohmmeter operating at a potential of 500 volts and having an accuracy range from 0 to 10 megohms. Failure to meet the requirements of 3.8 shall be cause for rejection.

4.7 Inspection of preparation for delivery requirements An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with 4.7.1. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:25Z
Check the source to verify that this is the current version before use.

with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

4.7.1 Classification of defects.

4.7.1.1 Packaging.

Major:
101 Closure of fiberboard container incorrect.
102 Fiberboard box not waterproof-sealed with tape (for level A).

4.7.1.2 Packing and marking.

Major:
101 Closure of shipping container incorrect.
102 Strapping missing when required.
103 Strapping not specified type and class, or incorrectly applied.
104 Gross weight of shipping container exceeds limit specified.
105 Case liner (if required) missing.
106 Markings not as specified.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Reels of belting and assembly tools shall be packaged individually in a fiberboard box conforming to PPP-B-636 (weather-resistant class) waterproof sealed with tape in accordance with the appendix thereto.

5.1.2 Level B. Unless otherwise specified (see 6.5), reels of belting and assembly tools weighing 25 pounds or less shall be packaged in a fiberboard box conforming to PPP-B-636, class domestic. Reels of belting and assembly tools weighing in excess of 25 pounds do not require unit packaging.

5.1.3 Level C. Reels of belting and assembly tools shall be packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:25Z
Check the source to verify that this is the current version before use.

5.2.1 Level A. Belting shall be packed in boxes conforming to any of the following specifications:

PPP-B-585, class 3	- Wood, wirebound
PPP-B-591 (overseas type)	- Fiberboard, wood-cleated
PPP-B-601 (overseas type)	- Wood, cleated-plywood
PPP-B-621 (class 2)	- Wood, nailed and lock-corner

Box closures and strapping shall conform to the box specification and the appendix thereto. Flat steel strapping shall be type I, class B.

5.2.2 Level B. Belting packaged as specified in 5.1 and unpackaged belting and assembly tools weighing in excess of 25 pounds, shall be packed in a snug-fitting fiberboard box conforming to PPP-B-636, class domestic. Gross weight of shipping containers shall not exceed 65 pounds, except when the weight of one belt and assembly tools exceeds this weight. When one unit exceeds 100 pounds in weight the shipping container shall conform to PPP-B-585, class 1, PPP-B-640, class 1, or PPP-B-601, domestic type.

5.2.3 Level C. The complete equipment shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking (see 6.2).

5.3.1 Military agencies. In addition to any special marking required, packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. In addition to any special marking required, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The belting is intended for field replacement of V-belts. Static conductive and nonsparking belting is intended for use when it is necessary to dissipate static electrical charges and for use in explosive atmospheres.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:25Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Size and quantity of belting required (see 1.2.2 and 3.5).

(c) When static conductive and nonsparking belting is required (see 3.8).

(d) Number of assembly tools, if required (see 3.10).

(e) Length of belting of preproduction sample, if different (see 4.3.1).

(f) Level of packaging and level of packing required (see 5.1 and 5.2).

(g) Marking required (see 5.3).

6.3 Preproduction sample. When a contractor has been the supplier of identical equipment in recent orders, the contracting officer may waive the requirement for a preproduction sample in the contract or order, and arrange for an accompanying reduction in price. However, the invitation for bids should include the requirement for a preproduction sample when the successful bidder is a manufacturer whose unit has not passed the preproduction test.

6.4 Oil for resistance test. The ASTM oil No. 1 used during the test of 4.6.2 may be obtained from:

Enjay Company, Inc.
15 W. 51st Street
New York, New York 10019
Attn: Mr. A. B. Boehm

6.5 Level B packaging Level B packaging (see 5.1.2) has been developed for use by the Army and Air Force for specific conditions specified in 6.2. For Navy procurements, details of level B packaging,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:25Z
Check the source to verify that this is the current version before use.

when required, should be developed at the time of procurement when all conditions are known.

MILITARY INTEREST:

Custodians:
- Army - ME
- Navy - YD
- Air Force - 82

Review activities:
- Army - ME
- Navy - YD
- Air Force - 82

User activities:
- Army - CE
- Navy - MC, SH, CG

Preparing activity:
- Navy - YD

CIVIL AGENCY INTEREST

Interested activities:
- GSA - FSS
- COM - NBS
- District of Columbia
- JUS
- AGR

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price of this specification cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:25Z
Check the source to verify that this is the current version before use.